

environments. Incineration is carried out both on a small and a large scale by industries using special incineration equipment under controlled conditions. A waste-to-energy plant is a waste management facility that combusts waste to produce energy.

e-Waste management involves both manual and large, intensive technical processes. e-Waste is first dismantled manually and then sent for further processing. For example, in a laptop, components like capacitors, memories, LCD screen and batteries are first manually removed and separated. Then, the remaining parts are sent for recycling through mechanical processing. The mechanical processing plant may include a crushing unit, shredders, as well as magnetic, eddy-current and air separators.

Burn what can't be recycled

Once waste materials reach landfills, they all look the same and cannot be recycled properly. To get maximum benefit from waste materials, these first need to be sorted properly, either from trash bins or from disposal or recycling stations. Materials collected from trash bins are sent to disposal stations for recycling, incineration or to landfills.

Incineration not only helps get rid of waste, it converts waste into heat and electricity. So, it is advisable to burn materials that cannot be recycled and get the benefits of usable energy and heat while reducing pollution and health hazardous elements. However, this needs to be done under controlled environments.

AI in e-waste management

AI is the new saviour of e-waste management and recycling. AI, machine learning, image processing and robotics have a bright future in the waste management sector.

Two ways to sort waste materials before recycling are either dropping the waste into separate trash bins or letting the trash bins sort themselves through an automatic system.

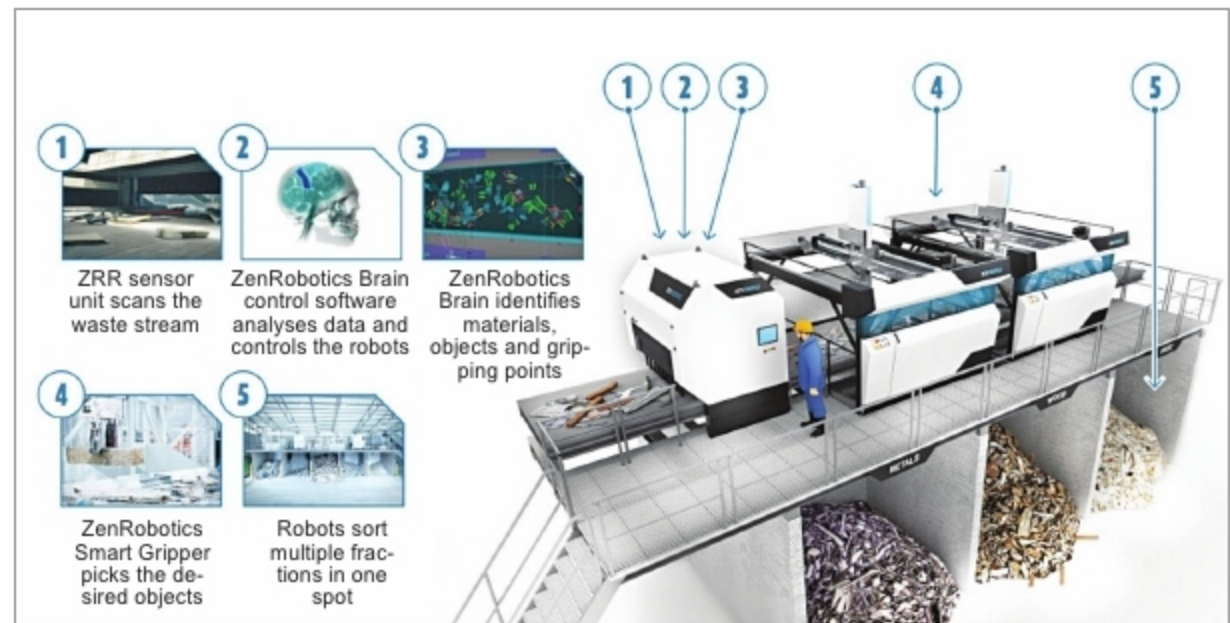


Figure 3: Robotic waste sorting system by ZenRobotics (Credit: zenrobotics.com)

If everything is dumped into one trash bin, sorting for recycling can be a tiresome task. To tackle this manually, different countries have different colour-coded dustbins for dumping different waste materials. But manual separation of waste materials or throwing into separate trash bins is somewhat confusing as human beings can make mistakes. People can get confused as to where to dump the waste, because they are not sure of the actual material of the packaging and devices. This is where AI comes in.

Employing AI in garbage sorting and disposal processes is a better method for smart recycling and waste management. Therefore AI and various sensors including RFID tags are used. Many intelligent dustbins have been developed that are equipped with AI programs and Internet of Things (IoT) sensors in the waste management sector. Multitasking and sorting AI robots have been developed for waste management that can sort tons of garbage on a daily basis.

In South Korea, RFID tags and sensors are used to collect garbage. A pneumatic garbage disposal system then reads these tags. The main computer, which stores all data, determines an appropriate method to dispose of the waste.

Chinese company Baidu and the United Nations Development Program are working to streamline the recycling of e-waste in China with a new app

that can help users sell their old electronics for cash.

Rubicon Global has a cloud-based, full-service waste management and recycling solution. AI and machine learning are incorporated into the solution to enhance human insight and improve waste diversion. The company has also developed an app that uses computer vision to generate a quick assessment of the waste stream, and a device is mounted on garbage trucks to visually analyse roadside dustbins.

As per a report from *mirror.co.uk*, Bin.E is the world's first intelligent bin that sorts waste materials. It uses a combination of sensors, image recognition and AI, and can recognise objects like glass, plastic or paper, and compress these before placing them in containers.

Developed by CleanRobotics, TrashBot robot is the first ever smart trash robot that uses AI to sort recyclables from landfill waste.

CP Manufacturing has a complete line of recycling equipment to ensure the highest recovery rates for virtually any material type, including commercial and industrial e-waste, food waste and green waste. L-VIS is a high-resolution colour camera sorting system from CP Manufacturing that provides accurate colour and size separation of small particles, and is suitable for electronic scrap, plastic flake, metal scrap, glass and other small particle segregation.